Bi-Weekly Status Report

Project Name: Code Sentinel – A Multi-Agent Code Review Platform
Report Period: 01 Nov 2026 – 16 Nov 2026
Prepared By: Sumit Goyal (Orchestration – on behalf of the team)
Date Submitted: 16/11/2026

Project Status Summary

Sprint/Milestone	Final report of the project. M7 (Test Plan and Test Cases) delivered; M8 (testing and deployment) complete with the application live at `codesentinel.scribly.fun`; final project report, deployment notes and README submitted; M9 (final presentation, 18 Nov) and M10 (peer review, 19 Nov) remain
% Complete (overall)	98%

Indicator	Current	Previous	Trend	Status
Overall Project	98%	88%	↑	**On Track**
Schedule	100%	95%	↑	**On Track**
Scope	100%	100%	→	**On Track**
Quality	868 tests (CI green)	863 tests (CI green)	↑	**On Track**
Team Contribution	100%	100%	→	**On Track**
Dependencies	100%	100%	→	**On Track**

Summary Notes:
This is the last bi-weekly report. The application was already feature-complete and in production at the end of the previous period, so these two weeks went to testing, evidence and the final submission. The Test Plan and Test Cases document (M7) was delivered on 2 Nov. A seed script now fills all 18 database tables with realistic, connected demo records, with its own tests, so every dashboard screen has data to show. A live check of the deployed platform on 11 Nov confirmed that the gateway, orchestrator and database are healthy and that unauthenticated calls, unsigned webhooks and the metrics endpoint are refused. The repository now carries the three final deliverables: a README describing the complete feature set and how frontend, backend and database fit together; deployment notes covering hosting, CI/CD and security configuration; and the 19-page final project report with screenshots of the live application. 9 commits in the period; 868 tests pass in CI on Node 22 and 24. What remains is the final presentation on 18 Nov and the peer review on 19 Nov.

Key Accomplishments in Two Weeks

- **Test Plan and Test Cases (M7)** – test strategy, unit, integration, API, database and non-functional cases, and the bug log, with every member writing the cases for their area (`7cef11e`)

- **Demo data in every table** – a seed script fills all 18 tables with two organizations, reviews in every state, clubbed findings, accepted and dismissed fixes, similar-issue links, devices and audit events; it can be re-run safely and is refused in production (`691452c`)
- **Seed data under test** – unit tests for the seed helpers and an integration test that seeds a fresh PostgreSQL twice and checks the counts and links (`72d295b`)
- **Live check of the deployed platform** – health of gateway, orchestrator and database; 401 for a missing token, an unsigned webhook and the metrics endpoint; security headers on the dashboard; results recorded in the deployment notes (`d105be5`)
- **Seeded-bug review checked against its answer key** – the live review of a demo pull request with planted security, logic, performance, style and documentation bugs (45 findings in 24 places, all 7 files covered, in 34 seconds) was compared with the list of planted bugs for each agent
- **Deployment notes** – hosting, public URLs, CI/CD, security configuration and monitoring in one document (`d105be5`)
- **README for the final application** – complete feature list, how frontend, backend, database and sign-in fit together, and the APIs and external services used (`567262e`)
- **Screenshots and brand logo** – ten screenshots of the live dashboard and of the review on GitHub, used in the README and the final report (`56904df`, `cd4cc7d`)
- **Final project report** – 19 pages: summary, system overview, technology stack, features, challenges and solutions, screenshots, testing and deployment (`b8baa38`)
- **Design notes brought up to date** – the large-diff and agent plans now describe whole-file review and the embeddings budget as built (`ecd34ef`)

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
M7 Test Plan and Test Cases	Nov 06	—	Done (Nov 02)	Delivered four days early; in the repository as a PDF
Live end-to-end tests on the deployed platform	Nov 14	—	Done by hand (Nov 11)	Run as a live check and a seeded-bug pull request rather than an automated suite; results in the deployment notes
Load and failure tests	Nov 14	—	Partly done	Agent timeouts, quota exhaustion and sleeping agents are covered by automated tests; no separate load run was made

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Requirements and design document updates	Nov 14	—	Changed	Deployment, monitoring and access rules are documented in the deployment notes, README and final report instead of re-issuing the earlier PDFs
Final demonstration preparation	Nov 14	Nov 18	In Progress	Demo data, seeded-bug repository and screenshots ready; presentation on 18 Nov
Final project report, deployment notes and README	Nov 16	—	Done (Nov 16)	All three in the repository; application live
Seventh bi-weekly report	Nov 16	—	Done	This report

Project Risks, Issues, & Dependency

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Risk (R-11)	M7 Test Plan not started, due 06 Nov	Medium	Ayush Soni	**Closed**: delivered on 2 Nov
Risk (R-10)	Free hosting sleeps idle services, so the first review after a quiet spell is slow	Medium	Nevil Nandasana	Agents are woken before each review and progress is shown live; before the final presentation the agents are warmed with a throwaway review
Risk (R-01)	LLM free-tier quotas shared by four LLM agents	Low	Sumit Goyal	**Mitigated**: quota reserved before each call, Gemini as overflow, Grafana alert when quota runs low; no quota error seen in the period

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Issue (I-04)	Render does not redeploy every service on its own after a push	Low	Vatsal Kamani	Services are redeployed by hand after a release and checked with the health endpoints; no code release is planned before the presentation
Issue (I-06)	No separate load run against the deployed platform	Low	Sumit Goyal	Accepted: free hosting limits make a load run unrepresentative; rate limits and quota planning are covered by automated tests
Risk (R-12)	A late change could break the live demo	Medium	Vatsal Kamani	The application is frozen from 16 Nov; only documents change until the peer review
Risk (R-09)	The requirements document is a PDF only	Low	Sumit Goyal	**Closed**: changes since the requirements are recorded in these reports, the deployment notes and the final report

Project Decisions

Decision	Rationale	Impact	Decided By	Date
Demo data comes from a seed script, never from production	Screens need realistic data for the demo and for testing, without touching live reviews	Seed runs on development databases only and is refused in production	Team	Nov. 04
Live testing by hand on a seeded-bug repository	An automated suite against free hosting would spend LLM quota and wake services on every run	One documented live check and one full review instead of a suite	Team	Nov. 10
Deployment notes in their own document	The README had grown too long to read as an introduction	README keeps a summary and links to `docs/DEPLOYMENT.md`	Team	Nov. 12

Decision	Rationale	Impact	Decided By	Date
Source code submitted as the repository archive	The archive holds only committed files, so no keys or environment files can be included	No hand-made archive to keep in step	Team	Nov. 14
Application frozen from 16 Nov	The final presentation and peer review should see the submitted version	Only documents change after this date	Team	Nov. 16

Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
Demo data in all 18 tables	Added	The earlier seed created only one organization, repository and user	Absorbed within the period	Team
Automated live end-to-end suite	Changed	Replaced by a documented live check and a seeded-bug review, to stay within free hosting and LLM limits	None	Team
Separate load run	Removed	Not representative on free hosting; limits and quota planning are tested automatically	None	Team
Re-issue of the requirements and design PDFs	Removed	Later changes are recorded in the deployment notes, README and final report	None	Team

Upcoming Work (Next Two Week's Plan)

- Final presentation and live demonstration, M9, 18 Nov (whole team; Moksh Mehta and Vatsal Kamani run the demo)
- Warm-up of the deployed services and a rehearsal review on the seeded-bug repository before the presentation (Nevil Nandasana, Tej Prakash Tak)
- Peer review, M10, 19 Nov (whole team)
- Submission of the final report and repository archive on Google Classroom (Sumit Goyal)

- Keeping the deployed services and monitoring running until the evaluation is over (Vatsal Kamani, Ayush Soni)

Team Member Contributions

Name: Sumit Goyal
Role: Orchestration – LangGraph workflow & LLM abstraction layer

Tasks Completed	Final project report; README for the final application (features, integration, APIs); large-diff and agent plans brought up to date; orchestrator and LLM test cases in the Test Plan; live check of the deployed platform; submission coordination; this report
Tasks In Progress	Final presentation; submission on Google Classroom
Contribution Type	Documentation / Testing / Architecture
Blockers	None
Hours/Capacity	212/200 hrs logged
Evidence	3 commits: `ecd34ef`, `567262e`, `b8baa38`; `README.md`, `docs/CodeSentinel_Final_Project_Report.pdf`

Name: Durgesh Khushlani
Role: Orchestration – Aggregation, severity ranking & Vector DB integration

Tasks Completed	Tests for the seeded demo data, including an integration test on a fresh PostgreSQL with pgvector; database package notes for the seed; aggregation, de-duplication and vector-context test cases in the Test Plan; check of clubbed findings and "seen before" links on the seeded-bug review
Tasks In Progress	Final presentation
Contribution Type	Testing / Development / Documentation
Blockers	None
Hours/Capacity	210/200 hrs logged
Evidence	1 commit: `72d295b`; `packages/db`; Test Plan integration and database cases

Name: Tej Prakash Tak
Role: Agent Services – Security Agent & Logic Agent

Tasks Completed	Security and Logic agent test cases and security non-functional cases in the Test Plan; planted security and logic bugs in the demo repository and checked the agents' findings against them; rate-limit and authentication checks on the live gateway; bug rota
Tasks In Progress	Rehearsal review before the presentation
Contribution Type	Testing / Security / Documentation
Blockers	None
Hours/Capacity	208/200 hrs logged
Evidence	Test Plan unit, API and security cases; review of the seeded-bug pull request on the demo repository (16 Security and 4 Logic findings)

Name: Nevil Nandasana
Role: Agent Services – Style Agent

Tasks Completed	Style Agent and sandbox test cases in the Test Plan; planted style bugs in the demo repository and checked the exact fixes; reliability cases for sleeping agents and timeouts; check of metrics and alert rules on Grafana Cloud; bug rota
Tasks In Progress	Warm-up of the deployed services before the presentation
Contribution Type	Testing / DevOps / Documentation
Blockers	None
Hours/Capacity	206/200 hrs logged
Evidence	Test Plan unit, reliability and monitoring cases; review of the seeded-bug pull request (26 Style findings)

Name: Parin Makwana
Role: Agent Services – Performance Agent & Documentation Agent

Tasks Completed	Performance and Documentation agent test cases in the Test Plan; planted performance and documentation bugs in the demo repository and checked the findings; sessions and signed-in devices test cases; proofreading of the final report and deployment notes; bug rota
Tasks In Progress	Final presentation
Contribution Type	Testing / Documentation
Blockers	None
Hours/Capacity	205/200 hrs logged
Evidence	Test Plan unit and API cases; review of the seeded-bug pull request (7 Performance findings); final report review

Name: Ayush Soni
Role: Backend & Database – API Gateway & PostgreSQL

Tasks Completed	Test Plan and Test Cases document, coordinating every member's cases; seed script filling all 18 tables with connected demo records; gateway, authentication and database test cases; check of migrations and row-level security on the live database
Tasks In Progress	Keeping the deployed services running through the evaluation
Contribution Type	Development / Testing / Documentation
Blockers	None
Hours/Capacity	212/200 hrs logged
Evidence	2 commits: `7cef11e`, `691452c`; `packages/db`, `docs/CodeSentinel_Test_Plan_and_Test_Cases.pdf`

Name: Vatsal Kamani
Role: Integrations & DevOps – Monorepo, GitHub App & Deployment

Tasks Completed	Deployment notes: hosting, public URLs, CI/CD, security configuration, monitoring and the live check; CI and deployment test cases in the Test Plan; check that CI gates every Render deploy; GitHub App webhook and Check Run checks on the live system
Tasks In Progress	Final demonstration; keeping the deployed services running
Contribution Type	DevOps / Documentation / Testing
Blockers	None
Hours/Capacity	209/200 hrs logged
Evidence	1 commit: `d105be5`; `docs/DEPLOYMENT.md`; green CI run on the final commit

Name: Moksh Mehta
Role: Frontend – Next.js dashboard

Tasks Completed	Screenshots of the live dashboard and of the review on GitHub for the README and final report; brand logo; interface and responsive-layout test cases in the Test Plan; pass over every dashboard screen with the seeded demo data; slides for the final presentation
Tasks In Progress	Final demonstration
Contribution Type	Design / Testing / Documentation
Blockers	None
Hours/Capacity	208/200 hrs logged
Evidence	2 commits: `cd4cc7d`, `56904df`; `docs/screenshots`, `docs/design/brand`